

Pre-Projection Admissible State Spaces for Relational Rotation

A Structural Note on Flow, Coarse-Graining, and Finite Observational Readout

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Abstract

Finite observational outcomes are often described as if they were the states of the underlying system. This note separates these two roles. We introduce an *admissible state space* M_{adm} as a pre-projection structural layer that specifies which states are allowed before a particular observation or coarse-graining is chosen. Dynamics is then represented by a vector field or flow on M_{adm} , while an observation protocol is represented by a map $\Pi: M_{adm} \rightarrow O$. The corresponding observational state space is naturally the quotient $M_{adm}/\sim_{\Pi}$, where $s_1 \sim_{\Pi} s_2$ when the protocol cannot distinguish the two states.

A planar relational-rotation example is used to make the separation explicit. Coordinates (X, Y) are treated only as one representation of an admissible state, not as binary variables. A continuous angular degree of freedom can be observed through binary, eight-class, or twelve-sector readouts without identifying any of those readouts with the dimension or ontology of the admissible space. We further show that common absolute rotation cancels from a relative-angle description, so an absolute background is unnecessary for the kinematics of internal relational rotation. The note deliberately does **not** claim that M_{adm} is a hidden-variable space, a new quantum state space, or an ontologically fundamental layer. Instead, it freezes a research step: finite readout, admissible dynamics, and deeper transition structure must be treated as distinct levels. The next problem is whether the admissible state space itself can be reconstructed from a more primitive transition relation.

Keywords: admissible state space; relational rotation; dynamical systems; flow; coarse-graining; quotient space; projection; aliasing; finite readout; symbolic dynamics; structural hypothesis.

中文摘要

有限观察结果常被直接当作系统本身的状态, 但二者并不必然相同。本文把这两个角色严格分开。我们定义一个投影前的 **允许态空间** M_{adm} , 用来描述在选定具体观察方式或粗粒化方式之前, 系统允许处于哪些状

态。动力学由 M_{adm} 上的向量场或流描述；观察协议则表示为映射 $\Pi: M_{adm} \rightarrow O$ 。若两个允许态在同一观察协议下给出相同结果，即 $\Pi(s_1) = \Pi(s_2)$ ，则它们在该协议下不可区分，并诱导出商空间 $M_{adm}/\sim_{\Pi}$ 。

本文用二维关系旋转作为最小工作例子。 (X, Y) 仅被视为允许态的一种坐标表示，**不等同于二进制 0/1，也不等同于量子八态或十二态**。同一个连续角自由度可以被观察为二值、八类或十二扇区读出，而这些有限读出不能反过来被解释为允许态空间的维数或本体。进一步地，在纯相对角描述中，两个对象同时进行的整体绝对旋转会消去，因此内部关系旋转的运动学不需要绝对背景。本文明确不主张 M_{adm} 是隐藏变量空间、新量子态空间或最终物理本体；本文只冻结当前研究链中的一个结构层：有限读出、允许态动力学和更深的转移结构必须分层处理。下一步问题是：允许态空间本身能否由更原始的状态转移关系重建。

1. Motivation and scope

A recurring source of confusion in multilevel descriptions is the silent identification of three different objects:

1. the set of states a model permits;
2. the dynamical rule that moves the system through those states; and
3. the finite labels returned by a particular observation.

These objects are generally not interchangeable. A binary detector may return only 0 or 1 while being driven by a continuous phase. A twelve-sector angular encoder may return one of twelve labels while the underlying angular configuration remains a point on the circle S^1 . The number of readout labels therefore does not, by itself, establish the dimension, cardinality, or ontology of the pre-observation state space.

This note formalizes a research step motivated by a longer chain of work on binary coarse-graining and multilevel observation [1]. The present step moves **upstream of the readout** and asks: what mathematical structure must exist before a finite observation can be defined?

The answer developed here is intentionally modest. We introduce an admissible state space, a dynamical flow, and an observation map, and we examine their separation using relational rotation as a benchmark. The components are standard mathematical objects from dynamical systems, symbolic dynamics, and quotient constructions [2-6]. The contribution of this note is not a claim that these ingredients are individually new. Rather, it is the explicit freezing of a structural hierarchy for a continuing research program, together with hypotheses and failure conditions that prevent different observational layers from being conflated.

1.1 Boundary statement

Boundary. This work does not claim that the admissible state space is the final ontology of nature; does not identify (X, Y) with binary variables; does not derive quantum mechanics from a classical planar state space; and does not claim that binary, eight-class, or twelve-sector readouts are new physical states. These are treated as distinct structural or observational levels unless an independent physical bridge is supplied.

2. Admissible state space

Definition 1: admissible state space

Let

$$M_{adm}$$

be a set, manifold, graph, or other state structure containing the states admitted by the model before a specific observation is chosen. A state is denoted

$$s \in M_{adm}.$$

The notation is deliberately representation-neutral. In one chart a state may be written as (X, Y) ; in another it may be represented by an angle θ ; in a discrete model it may be a node S_i . Coordinates are not identified with ontology.

For the simplest planar rotation benchmark, one may write

$$X = A \cos \theta, Y = A \sin \theta,$$

or equivalently

$$\psi = X + iY = A e^{i\theta}.$$

When A is fixed and positive, the configuration reduces to the circle S^1 . When A varies, the admissible set may occupy an annulus, disk, cylinder, or another structure depending on the model.

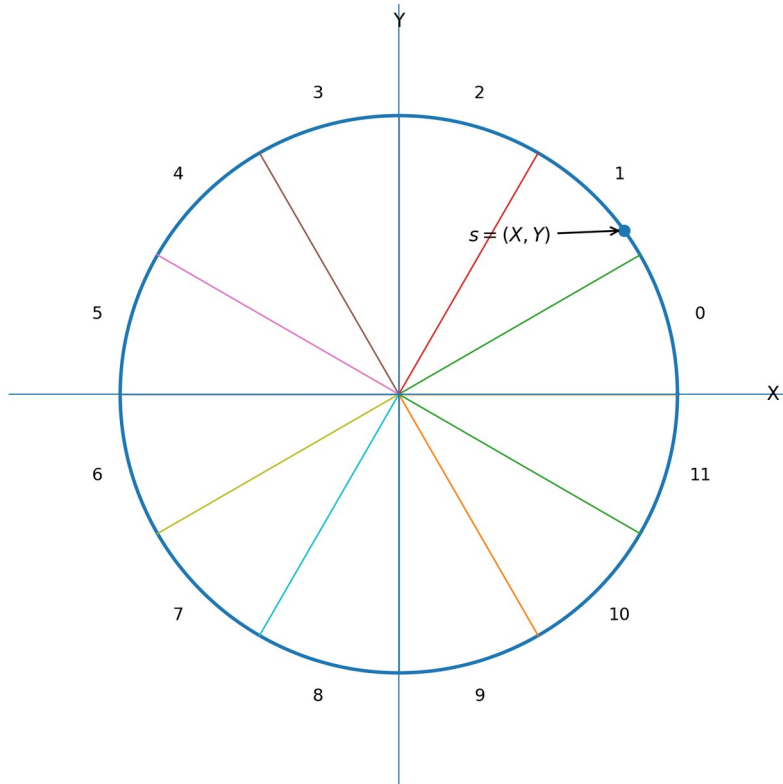


Figure 1. A continuous admissible angular state can be represented in (X, Y) coordinates and partitioned into twelve observational sectors. The sectors are a readout choice, not twelve underlying ontological states.

2.1 What (X, Y) means - and what it does not mean

The pair (X, Y) should be read as a coordinate representation of a state that supports a continuous orientation relation. It is **not** the same object as the final binary outcome 0/1. In particular,

$$X \neq 0, Y \neq 1.$$

The same state space can support many possible observation maps. Therefore the finite alphabet used for readout is logically downstream of the admissible state space.

3. Dynamics: vector field, flow, and trajectory

An admissible space alone does not produce motion. Dynamics requires a rule.

Definition 2: dynamical rule

For a smooth state space, let

$$F: M_{adm} \rightarrow T M_{adm}$$

be a vector field. The state evolves according to

$$\dot{s} = F(s).$$

For planar uniform rotation,

$$F(X, Y) = (-\omega Y, \omega X),$$

so that

$$\dot{X} = -\omega Y, \dot{Y} = \omega X.$$

Using

$$J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, J^2 = -I,$$

this becomes

$$\dot{s} = \omega J s,$$

with finite evolution

$$R(\theta) = e^{\theta J}.$$

Definition 3: flow

The vector field generates a flow

$$\Phi_t: M_{adm} \rightarrow M_{adm},$$

with

$$\Phi_t(s_0) = s(t).$$

A particular realized path

$$\gamma(t) = \Phi_t(s_0)$$

is a trajectory or orbit.

This yields four distinct objects:

Object	Mathematical role	Question answered
M_{adm}	admissible state structure	What states are allowed?
F	local dynamical rule	In which direction may a state change?
Φ_t	global flow	How does the state evolve over an interval?
$\gamma(t)$	realized trajectory	Which path is actually followed?

These levels should not be collapsed into a single word such as “field” or “state.”

4. Observation as projection or coarse-graining

Definition 4: observation map

Let

$$\Pi : M_{adm} \rightarrow O$$

be an observation, readout, or coarse-graining map. The observed symbol is

$$y = \Pi(s).$$

The codomain O may be finite even when M_{adm} is continuous.

Examples include

$$\Pi_2 : M_{adm} \rightarrow \{0, 1\},$$

$$\Pi_8 : M_{adm} \rightarrow \{0, 1, \dots, 7\},$$

and

$$\Pi_{12} : S^1 \rightarrow Z_{12}.$$

For a twelve-sector angular encoder,

$$\Pi_{12}(\theta) = k \text{ iff } 2\pi k/12 \leq (\theta \bmod 2\pi) < 2\pi(k+1)/12, \quad k \in \{0, \dots, 11\}.$$

The boundary $2\pi \equiv 0$ means that “12” is the completion of a cycle, not a thirteenth angular state.

Proposition 1: every deterministic readout induces an observational quotient

Define

$$s_1 \sim_{\Pi} s_2 \Leftrightarrow \Pi(s_1) = \Pi(s_2).$$

Then $\sim_{\Pi}$ is an equivalence relation and the observational state space is naturally represented by the quotient

$$O_{\Pi} \cong M_{adm} / \sim_{\Pi}$$

when the readout labels correspond one-to-one with equivalence classes.

Proof. Equality of outputs is reflexive, symmetric, and transitive. Therefore the preimages of the readout labels partition M_{adm} . Each partition element is an equivalence class. $\square$

The proposition is elementary, but its conceptual consequence is important: a finite observed alphabet is a partition of the admissible structure, not evidence that the admissible structure itself contains only that many states.

5. Aliasing and information loss

Definition 5: observational aliasing

Two distinct admissible states are aliased under Π when

$$s_1 \neq s_2, \Pi(s_1) = \Pi(s_2).$$

Aliasing is therefore not an error in the underlying state. It is a property of the observation protocol relative to the state space.

If an intervention u changes the observation or dynamics, write the protocol as Π_u . A pair is operationally separable by the intervention family U if

$$\exists u \in U : \Pi_u(s_1) \neq \Pi_u(s_2).$$

This gives a direct bridge from coarse-graining to active model discrimination: an intervention is useful precisely when it breaks an observational equivalence that was present under the baseline protocol.

6. Relational rotation without an absolute background

The clock-face intuition suggests a background, anchor, and pointer. For a mechanical clock those are useful physical components. At the kinematic level, however, part of this structure can be removed.

Consider two oriented states with angles θ_a and θ_b . Define their relative orientation

$$\Delta\theta = (\theta_b - \theta_a) \bmod 2\pi.$$

A common absolute rotation by α gives

$$\theta_a \mapsto \theta_a + \alpha, \theta_b \mapsto \theta_b + \alpha.$$

Hence

$$\Delta\theta \mapsto (\theta_b + \alpha) - (\theta_a + \alpha) = \Delta\theta.$$

Proposition 2: common absolute orientation is redundant for internal relative-angle kinematics

Any observable that depends only on $\Delta\theta$ is invariant under the diagonal action

$$(\theta_a, \theta_b) \mapsto (\theta_a + \alpha, \theta_b + \alpha).$$

Therefore an absolute angular background is not required to represent the internal relative orientation of the two-state system.

This statement is purely kinematic. A real physical system may couple to an external field, laboratory axis, gravity vector, crystal lattice, or other reference that makes absolute orientation relative to that environment operationally relevant. The proposition only states what is removable when the observables are relational.

A convenient coordinate-free representation is

$$C = \cos \Delta \theta, S = \sin \Delta \theta,$$

so that the relative state

$$q = (C, S) \in S^1.$$

If the two orientations rotate with rates ω_a and ω_b , then

$$\dot{\Delta \theta} = \omega_b - \omega_a \equiv \Omega,$$

and

$$\dot{q} = \Omega J q.$$

Only the relative rate Ω remains.

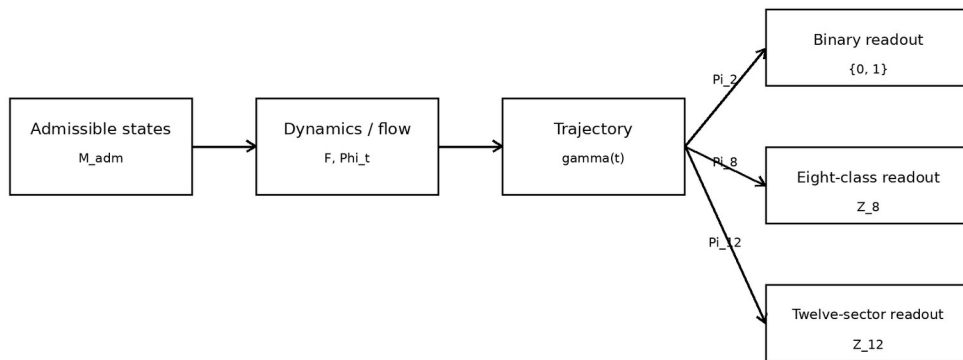
7. Finite readouts are different effective levels

A major hypothesis of this research chain is that different finite descriptions should be treated as distinct **effective observational spaces**, not automatically as different ontological layers.

For example, the same angular trajectory could be observed through:

- a binary partition Π_2 ;
- an eight-class partition Π_8 ;
- a twelve-sector partition Π_{12} .

These are different quotient structures. They may preserve different amounts of information and can have different aliasing patterns.



Different observation maps induce different observational quotients.

Figure 2. Structural hierarchy used in this note. The admissible state space and its flow precede any chosen finite readout. Different observation maps produce different effective alphabets and different equivalence classes.

Proposition 3: readout cardinality does not determine admissible-state dimension

For any integer $K \geq 2$, a continuous circle admits a deterministic K -sector map

$$\Pi_K: S^1 \rightarrow Z_K.$$

Therefore the existence of a K -symbol readout does not imply that the underlying configuration space has K states or dimension K .

The converse also fails: a low-dimensional state space can support arbitrarily fine finite partitions. Hence one must not infer ontological multiplicity from readout multiplicity alone.

8. Frozen hypotheses for the next research step

The present note deliberately separates formal results from hypotheses.

H1. Pre-projection richness hypothesis

A finite readout may arise as a coarse-graining of a richer admissible state structure:

$$M_{adm} \xrightarrow{\Pi} O.$$

This is mathematically possible and common. The physical question is whether a specific proposed M_{adm} is independently motivated and operationally distinguishable.

Failure condition: if all experimentally accessible predictions are completely captured by the finite readout model and no admissible-state refinement changes any possible task, then the richer structure has no established physical status.

H2. Relational-rotation hypothesis

For observables internal to a rotational subsystem, the physically relevant quantity may be relative orientation and relative rate rather than an absolute background angle.

Failure condition: an experimentally relevant coupling to an external reference makes the common rotation observable.

H3. Multi-readout non-equivalence hypothesis

Binary, eight-class, twelve-sector, and other finite readouts should be treated as distinct observational quotients unless a theorem establishes an equivalence between them.

Failure condition: a specific model proves that two readouts are related by an invertible recoding on the task-relevant state set, in which case they carry the same operational information for that task.

H4. Further-reducibility hypothesis

The admissible state space may itself be an effective representation rather than the deepest structure. A more primitive object may consist of states and allowed transitions,

$$S_i \rightsquigarrow S_j,$$

from which continuous geometry and flow emerge only at a higher descriptive level.

This hypothesis is intentionally left unresolved here. It defines the next step in the chain.

9. Relation to standard frameworks

The mathematical ingredients used here are standard:

- state spaces, vector fields, flows, and periodic orbits belong to classical dynamical systems [2,3];
- partitions of continuous dynamics into finite symbol sequences are central to symbolic dynamics [4];
- quotient constructions formalize identification under equivalence relations;
- predictive-state and computational-mechanics programs similarly distinguish histories that are operationally equivalent for future prediction [5,6];
- operator-based evolution provides another representation of dynamical structure, including the Koopman viewpoint [7].

The note therefore does not claim novelty for state spaces, flows, partitions, quotients, or relative coordinates individually. Its role is to impose a disciplined structural separation on a particular research chain and to freeze testable hypotheses before moving to a deeper level.

10. Why this layer is worth freezing even if it is not fundamental

A research chain can remain useful when an intermediate layer is later shown not to be fundamental. The present layer is designed to survive such a revision.

Suppose future work establishes a deeper transition substrate T . Then one may have

$$T \rightarrow M_{adm} \rightarrow O_{\Pi}.$$

The current note would remain valid as an effective intermediate representation, provided the mapping from T to M_{adm} is specified.

This motivates a versioned research protocol:

1. freeze each structural level with explicit assumptions;
2. distinguish theorem, representation, hypothesis, and empirical evidence;
3. define the bridge to the next level;
4. permit later levels to refine or supersede ontology without rewriting the historical chain.

11. Minimal worked example

Consider a uniform phase

$$\theta(t) = \omega t + \theta_0 \pmod{2\pi}.$$

The admissible configuration is S^1 . Define three observation maps:

$$\Pi_2(\theta) = \begin{cases} 0, & 0 \leq (\theta \bmod 2\pi) < \pi, \\ 1, & \pi \leq (\theta \bmod 2\pi) < 2\pi, \end{cases}$$

$$\Pi_8(\theta) = k \text{ iff } 2\pi k/8 \leq (\theta \bmod 2\pi) < 2\pi(k+1)/8, \quad k \in \{0, \dots, 7\}.$$

and

$$\Pi_{12}(\theta) = k \text{ iff } 2\pi k/12 \leq (\theta \bmod 2\pi) < 2\pi(k+1)/12, \quad k \in \{0, \dots, 11\}.$$

All three readouts are generated by the same continuous trajectory, but their equivalence classes differ. In particular, Π_2 merges half of the circle into each output, while Π_{12} retains a finer angular partition. None of these facts imply that the underlying dynamics contains two, eight, or twelve ontological states.

This example captures the central distinction:

$$\text{admissible state} \neq \text{observed label}.$$

12. Open problems and next death gates

The next stage should not add more finite labels. It should attempt to remove the state-space representation itself.

12.1 Pure relational reconstruction

Can all task-relevant rotational observables be reconstructed from relational variables alone, with no absolute background coordinate?

12.2 Transition-relation reconstruction

Can a continuous admissible space and its generator be recovered from only a set of states and an allowed-transition relation?

$$\{s_i\}, \rightsquigarrow \Rightarrow M_{adm}, F?$$

A useful result would be a necessary-and-sufficient condition for when such a transition structure admits an effective continuous rotational representation.

12.3 Discrete versus continuous substrate

Can a finite or countable transition system produce an exact $U(1)$ rotational degree of freedom, or only approximate it in a limit? The distinction between exact continuous symmetry and finite cyclic symmetry C_N is a critical death gate.

12.4 Operational bridge

A deeper structure becomes physically meaningful only if it changes a measurable probability, response, resource, or intervention task. Without such a bridge, it remains a representation or hypothesis generator.

13. Conclusion

This note freezes one intermediate level in a downward structural decomposition. The central object is not a binary variable and not a finite “hidden” alphabet. It is an admissible state structure M_{adm} on which dynamics acts before an observation map compresses the state into a finite readout.

For relational rotation, (X, Y) provides one convenient representation, but the coordinates themselves are not fundamental. A common absolute rotation can be removed from purely relational observables, leaving relative phase

and relative rate. Binary, eight-class, and twelve-sector descriptions can then be treated as different coarse-grained observation spaces rather than the same physical layer.

The present layer is intentionally provisional. Its purpose is to preserve a continuous research chain while keeping the boundary between mathematics and ontology explicit. The next step is to ask whether even M_{adm} can be eliminated in favor of a more primitive transition structure.

The resulting research direction can be summarized as

transition structure ? $\rightarrow M_{adm} \rightarrow$ flow $\rightarrow$ observation quotient $\rightarrow$ finite readout

with the question mark marking the next unresolved layer.

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Version note

v0.1 (2026-09-01): First frozen formulation of the pre-projection admissible-state layer. This version explicitly separates admissible states, dynamics, observation quotients, and finite readouts; introduces the relational-rotation benchmark; and defines transition-relation reconstruction as the next research death gate.